

DEVASHISH SONAWANE

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PROFESSIONAL SUMMARY

Healthcare Data Analyst and Biotech Researcher with 2+ years of industry experience in clinical informatics, predictive analytics, data engineering, genetics, and toxicology. Python, R, SQL, Power BI, Tableau, DICOM, and HIPAA workflows.

EDUCATION

University of Southern California, Viterbi School of Engineering, Los Angeles, CA

M.S. Healthcare Data Science | GPA: 3.91 / 4.00

Aug 2023 – May 2025

Courses: Machine Learning, Data Science at Scale, Data Visualization, Medical Imaging Systems, Clinical Informatics, Foundations of Data Mining, Analysis of Algorithms, Security & Privacy

D.Y. Patil University, School of Biotechnology & Bioinformatics, Navi Mumbai, India

M.Tech Biotechnology & Bioinformatics | GPA: 3.97 / 4.00

Aug 2021 – Jul 2022

B.Tech Biotechnology & Bioinformatics | GPA: 3.79 / 4.00

Aug 2017 – Jul 2021

Courses: Biostatistics, Genetics, Sequence & Structure Analysis, RNA-seq, BLAST, Pharmacokinetics, Nanobiotechnology, Immunology, Computational Drug Modeling & Design, Phyloinformatics, Structural Bioinformatics, Epidemiology

PROFESSIONAL EXPERIENCE

Healthcare Data Analyst

Mysti's Adult and Family Services

Aug 2025 – Present

- Designed and deployed a **Power BI** dashboard to track grievance case **KPIs**, employee performance metrics, and service utilization trends across **Medi-Cal** and **CalAIM** programs, enabling real-time visibility into care gaps.
- Extracted, queried, and audited clinical data from **JIVA** and **Ensora Health EMR** for CCAH; performed **ETL** operations to ensure data quality and integrity in compliance with **HIPAA**, Medi-Cal, and CalAIM standards.
- Analyzed **population health trends** and **patient outcomes** across grievance case data, surfacing recurring care gaps and quality improvement opportunities that informed process changes.

Data Science Research Intern

University of Southern California

Mar 2024 – May 2025

- Architected a clinical decision support tool for radiation therapy using **Python**, **SQL**, and clustering algorithms to help oncologists select safe radiation dosages by ranking spatially similar tumor cases from **DICOM** CT scans.
- Designed and deployed a **HIPAA-compliant MySQL** database to store anatomical and dosimetric data from **DICOM** RT files, supporting clinical querying and KPI monitoring for radiotherapy planning.
- Tested performance by engineering phantom datasets with known dissimilarity, confirming a strong **0.7 correlation**.
- Containerized the full **ETL** pipeline using **Docker**, reducing compute time and improving output consistency; applied model versioning and pipeline automation to ensure reproducibility in compliance with **IRB** protocols.

Teaching Assistant

University of Southern California

Aug 2024 – Dec 2024

- Supported 30+ graduate students through **Python**, **R**, and **MATLAB** lab sessions covering biostatistics, DICOM data handling, and medical imaging, grading coursework and addressing student questions.

Graduate Student Ambassador

University of Southern California

Mar 2024 – May 2025

- Represented the USC MS Healthcare Data Science program at 10+ recruitment events, directly advising 100+ prospective students on curriculum, research opportunities, and career pathways in health data science.

Data Analyst

Mega LifeSciences, Mumbai, India

Jan 2023 – Jul 2023

- Developed demand-forecasting models in **Python** using time-series analysis and regression on pharmaceutical and supply chain datasets, contributing to an estimated **~20% reduction** in operational costs.
- Consolidated regional sales, supply chain, and marketing datasets into unified **SQL** schemas, enforcing data governance standards and reducing manual reporting time by **~30%**.
- Built automated **Power BI** dashboards surfacing real-time KPIs across drug supply, sales performance, and marketing ROI for executive stakeholders, replacing manual weekly reporting cycles.

Research Data Analyst

ICAR – Central Institute of Fisheries Education

Dec 2021 – Dec 2022

- Optimized nanoparticle synthesis protocols by designing and testing a novel stabilizing method for 3 conjugated nanoparticle variants, increasing extraction yield by **260%** and reducing material waste.

- Conducted **genotoxicity** analysis on fish embryos by integrating **multi-omics** datasets and applying regression analysis in **SAS**, **STATA**, and **R** to identify key toxicity predictors.
- Managed a live fish research facility of **4,000+** specimens by optimizing breeding protocols, environmental parameter monitoring, and data-driven resource allocation, reducing specimen mortality rate by **70%**.
- Contributed to **DNA barcoding** genomic studies across multiple fish species by conducting molecular data extraction, quality control, and training 3 junior researchers in **GCP**-compliant laboratory protocols.

Tissue Engineering Research Intern

BioLim Centre for Science and Technology

Aug 2021 – Sep 2021

- Developed research designs for cultured meat production by synthesizing 50+ peer-reviewed sources on tissue engineering, **pharmacokinetics** of growth factors, and cellular biology to evaluate scaffold biocompatibility and viability.
- Streamlined patent writing and research publication workflows by coordinating literature mining, data synthesis, and submission processes for client research projects, improving turnaround efficiency.

Data Analyst

Save The Paws, Animal Welfare NGO

Jun 2021 – Feb 2022

- Applied **Python**-based predictive analytics to social media KPI data, boosting follower growth by **~40%**.
- Built interactive **Tableau** dashboards to visualize fundraising performance, adoption trends, and campaign ROI.

KEY PROJECTS

Radiation Therapy Decision Support System | Python, SQL, Docker, DICOM-RT, MySQL, Clustering

May 2025

- Designed and validated a **clinical AI** tool that processes **DICOM RT** imaging data to generate dosage recommendations by ranking spatially similar tumor/OAR cases using **Gower's distance clustering**.

Breast Tumor Classification | Python, Scikit-Learn, SVM, Monte Carlo

Aug 2024

- Trained **L1-penalized SVM** and semi-supervised classification models on the UCI Breast Cancer Wisconsin dataset to detect malignancy, evaluating performance using **ROC-AUC**, **Monte Carlo** simulation, and confusion matrices.

Activity Recognition & Classification | Logistic Regression, L1 Regularization, Cross-Validation

Oct 2024

- Extracted time-domain features from **AReM** multisensor time-series data and applied **L1-regularized logistic regression** with cross-validation to build an interpretable multi-class activity detection model.

Spatiotemporal Pattern Recognition | Apache Tika, Solr, D3.js, NLP, Docker, SMQTK, API Pipeline

Dec 2024

- Built interactive **D3.js** geospatial visualizations by integrating multiple datasets and analyzing 5,500+ sightings spanning over 200 years using **NLP**, entity detection, and **geo-topic parsing** under Dr. Chris Mattmann (CTIO, NASA JPL).

Breathe Better: Environmental Health Dashboard | React, D3.js, Deck.gl, Geospatial Analytics

Feb 2025

- Built an interactive public health dashboard using **React**, **D3.js**, and **Deck.gl** to visualize global pollution sources and linked health outcomes, integrating **WHO**, **EPA**, and **PurpleAir** datasets with dynamic maps and Sankey diagrams.

TECHNICAL SKILLS

Programming: Python, R, SQL, SAS, STATA, HTML, CSS, Linux/Unix, Perl, MATLAB, NoSQL, D3.js

Machine Learning & Analytics: Scikit-Learn, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, ggplot2, Plotly, NLTK, spaCy, Regression Analysis, Survival Analysis, A/B Testing, Time-Series Forecasting, Clustering, SVM, Random Forest, NLP, Semi-Supervised Learning

Healthcare & Clinical Informatics: DICOM, PACS, EHR (Epic), HIPAA Compliance, IRB Protocols, GCP, ICD-10, CPT, LOINC, HL7/FHIR, ETL, Data Quality, Population Health Analytics, Patient Outcomes Research, Oncology Analytics, Clinical Decision Support, Medical Imaging, Clinical KPI Tracking, Cohort Studies, Epidemiology

Bioinformatics & Biology: Sequence & Structure Analysis, RNA-seq, BLAST, DNA Barcoding, Multi-Omics, Phyloinformatics, Structural Bioinformatics, Pharmacokinetics, Genomics, Proteomics, Nanobiotechnology, Genotoxicity Assays, Drug Modeling & Design, Immunology, Molecular Biology

Business Intelligence: Power BI, Tableau, D3.js, Excel, Dashboard Design, KPI Reporting, EDA, Data Wrangling

Databases: MySQL, PostgreSQL, MongoDB, Google BigQuery

Cloud & DevOps: AWS, GCP, Microsoft Azure, Docker, Git, Jupyter, Apache Tika

Coursework: Kubernetes, PySpark, SAP, SMQTK, Solr, SPSS

CERTIFICATIONS

- IBM Certified NoSQL and DBaaS Developer, IBM
- Google Data Analytics Professional Certificate, Google
- Python for Everybody, University of Michigan

ACHIEVEMENTS & LEADERSHIP

- **1st Place, Roblox × USC Open Alpha Game Jam (UX & Accessibility, Apr 2024):** Built 'Mystical Memories: The Genie's Challenge' in Roblox Studio within 48 hours, earning recognition for User Experience and Accessibility.
- **Published Author:** Published article 'Use of Telomerase Catalytic Protein Component to Stop Effects of Ageing' in TORCH Magazine, with a co-authored nanoparticle genotoxicity research manuscript currently under peer review.